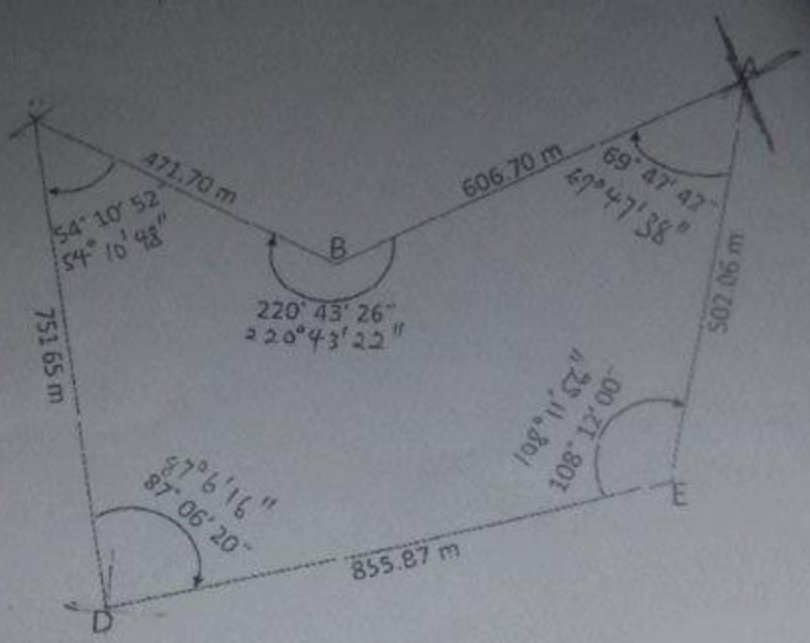


Fig. 1: Closed Loop Traverse ABCDEA

Fig. 1 shows a diagram of a closed loop traverse ABCDEA conducted by students of KNUST during their practical task. The diagram is not drawn to scale. The unadjusted left hand angles at the stations and the distances between the stations are as indicated on the diagram.

The coordinates of Station **A** are **N 1000.00 m** and **E 2000.00 m,** and the forward bearing of **AE** is **191° 29□ 20"**. Complete Tables 3 and 4, or otherwise, and answer **questions 53 – 68.**

Table 3: Computation of Forward Bearings of Traverse Legs

Station		Back Bearing of Previous Leg	Adjusted Left Hand Angle	Forward Bearing	Traverse Leg
From	To				
E	A			11° 29' 20"	EA
A	B	191° 29' 20"	69°47'38"	261°16'58"	AB
B	C	81°16'58"	220°43'22"	302°0'20"	BC
C	D	120°0'20"	54°10'48"	176°11'08"	CD
D	E	356°11'8"	87°6'16"	83°17'24"	DE
E	A	263°17'24"	108°11'56"	11°29'20"	EA

57. The back bearing of line DE is
(a) 263° 17' 24"
(b) 263° 17' 28"
(c) 83° 17' 24"
(d) 83° 17' 28"
(e) None of the above

58. The unadjusted latitude of line AB is
(a) + 91.95 m
(b) - 91.95 m
(c) + 599.69 m
(d) - 599.69 m
(e) None of the above

59. The unadjusted departure of line CD is
(a) + 50.00 m
(b) - 50.00 m
(c) + 749.98 m
(d) - 749.98 m
(e) None of the above

60. What is the total error in latitudes?
(a) + 0.32 m
(b) - 0.32 m
(c) +0.07 m
(d) - 0.07 m
(e) None of the above

61. What is the total error in departures?
(a) + 0.32 m
(b) - 0.32 m
(c) +0.07 m
(d) - 0.07 m
(e) None of the above

62. Compute the linear misclosure of the traverse corrected to 3 decimal places.
(a) 0.390 m
(b) 0.250 m
(c) 0. 022m
(d) 0.328 m
(e) None of the above

63. What is the fractional misclosure of the traverse corrected to the nearest hundred?

(a) 1 : 8,200
(b) 1 : 9,700
(c) 1 : 12,800
(d) 1 : 9,800
(e) None of the above

64. Using Bowditch method of adjustment, the corrections to be applied to the latitude and departure line CD are

(a) +0.02 m and +0.07 m, respectively
(b) - 0.02 m and -0.07 m, respectively
(c) +0.02 m and -0.07 m, respectively
(d) +0.07 m and -0.02 m, respectively
(e) -0.07 m and +0.02 m, respectively

65. What are the adjusted northing and easting coordinates of station B?

(a) **N** 908.04 m and **E** 1400.25 m
(b) **N** 1400.25 m and **E** 908.04 m
(c) **N** 1091.94 m and **E** 2599.63 m
(d) **N** 2599.63 m and **E** 1091.94 m
(e) None of the above

66. Compute the adjusted coordinates of station C.

(a) **N** 508.01 m and **E** 1900.05 m
(b) **N** 1900.05 m and **E** 508.01 m
(c) **N** 408.03 m and **E** 1050.13 m
(d) **N** 1158.03 m and **E** 1000.20 m
(e) None of the above

67. Compute the adjusted coordinates of station D.

(a) **N** 908.04 m and **E** 1400.25 m
(b) **N** 508.01 m and **E** 1900.05 m
(c) **N** 408.03 m and **E** 1050.13 m
(d) **N** 1158.03 m and **E** 1000.20 m
(e) None of the above

68. Compute the adjusted coordinates of station E.

(a) **N** 908.04 m and **E** 1400.25 m

(b) **N** 508.01 m and **E** 1900.05 m

(c) **N** 408.03 m and **E** 1050.13 m

(d) **N** 1158.03 m and **E** 1000.20 m

(e) None of the above

69. Which of the following conditions must be satisfied when applying Simpso areas of irregular figures?

i. The number of offsets must be even.

ii. The offsets must be at regular intervals.

iii. The number of strips must be even.

(a) I only

(b) I and II only

(c) I and III only

(d) II and III only

(e) All of the above

70. The coordinates of a triangular parcel ABC are given by A(650.000 m 846.663 mN), and C(593. 349mE, 827.738mN). Compute the area o

(a) 196.456 m^2

(b) 119.332 m^2

(c) 169.456 m^2

(d) 149.337 m^2

(e) None of the above

50. What is the misclosure of the levelling task?
(a) – 0.003 m
(b) +0.003 m
(c) +0.009 m
(d) -0.009 m
(e) None of the above.

51. Calculate the correction to be applied to the initial reduced level of Peg 5.
(a) - 0.003 m
(b) +0.002 m
(c) - 0.002 m
(d) +0.003 m
(e) None of the above.

52. The adjusted reduced level of Peg 6 is
(a) 100.098 m
(b) 100.096 m
(c) 100.102 m
(d) 100.195 m
(e) None of the above.

53. Which of the following statements is not true?
(a) The sum of the backsights is equal to 5.660 m.
(b) The sum of the intersights is equal to 7.795 m
(c) The number of foresights is 3.
(d) The adjusted reduced level of Peg 3 is 100.124 m.
(e) None of the above.

44. What is the value of the second height of plane of collimation?
(a) 101.612 m
(b) 99.388 m
(c) 99.521 m
(d) 101.588 m
(e) None of the above.

45. Which of the following values is equal to the third backsight reading?
(a) 1.988 m
(b) 2.901 m
(c) 1.715 m
(d) 2.091 m
(e) None of the above.

46. At which instrument station was the greatest number of staff readings taken?
(a) First
(b) Second
(c) Third
(d) Fourth
(e) None of the above.

47. Which of the following sets of readings were taken from the same instrument position?
(a) 1.957, 1.762, 1.859, 2.670
(b) 1.715, 2.091, 1.862, 1.988
(c) 1.612, 2.312, 1.715, 2.091
(d) 1.988, 1.957, 1.762, 1.859, 2.670
(e) None of the above

48. What is the initial or unadjusted reduced level of Peg 1?
(a) 101.612 m
(b) 99.897 m
(c) 99.300 m
(d) 100.126 m
(e) None of the above

49. What is the initial or unadjusted reduced level of Peg 4?
(a) 101.612 m
(b) 99.897 m
(c) 99.300 m
(d) 100.126 m
(e) None of the above

During a levelling operation, the following staff readings (recorded in metres) were taken consecutively: 1.612, 2.312, 1.715, 2.091, 1.862, 1.988, 1.957, 1.762, 1.859 and 2.670. The first staff position was on a bench mark A whose reduced level is 100.000 m and the last staff reading was taken when the staff was placed on a bench mark B of reduced level 99.290. The position of the levelling instrument was changed after the third and sixth staff readings. Complete Table 2 below, adjust the reduced levels and use it to answer questions **42 – 53**.

Table 2: Levelling Reduction by Height of Plane of collimation Method

Backsight	Intersight	Foresight	Height of Collimation	Initial Reduced Level	Remarks
1.612			101.612	100.000	BM A(100.000 m)
	2.312				Peg 1
2.091		1.715	101.988		Peg 2
	1.862				Peg 3
1.957		1.988	101.957		Peg 4
	1.762				Peg 5
	1.859				Peg 6
		2.670			BM B (99.290 m)

42. How many instrument stations were used in all?

(a) 10

(b) 5

(c) 3

(d) 2

(e) None of the above.

43. The number of intersights minus the number of backsights is equal to

(a) 1

(b) 3

(c) 4

(d) 7

(e) None of the above.

37. What is the bearing of line QL?
(a) 123° 49' 55"
(b) 333° 54' 45"
(c) 153° 54' 45"
(d) 153° 24' 45"
(e) None of the above.

38. The latitude of line QL is
(a) -56.651 m
(b) -27.738 m
(c) 56.651 m
(d) 27.738 m
(e) None of the above.

39. The departure of line QD is
(a) -46.663 m
(b) -84.535 m
(c) 46.663 m
(d) 84.535 m
(e) None of the above.

40. The northing and easting coordinates of D are respectively
(a) 593.349 m and 846.663 m.
(b) 565.465 m and 827.738 m.
(c) 565.465 m and 846.663 m.
(d) 593.349 m and 827.738 m.
(e) None of the above.

41. The northing and easting coordinates of L are respectively
(a) 593.349 m and 846.663 m.
(b) 565.465 m and 827.738 m.
(c) 565.465 m and 846.663 m.
(d) 593.349 m and 827.738 m.
(e) None of the above.

31. The height of the instrument is
(a) 1.450 m
(b) 1.981 m
(c) 1.497 m
(d) 47.580 m
(e) None of the above.

32. The reduced level at the base of corner D is
(a) 51.703 m
(b) 48.917 m
(c) 45.701 m
(d) 52.359 m
(e) None of the above.

33. Compute the reduced level at the base of corner L.
(a) 51.703 m
(b) 48.791 m
(c) 46.357 m
(d) 52.359 m
(e) None of the above.

34. The horizontal distance of line QL is
(a) 96.559 m
(b) 65.456 m
(c) 63.077 m
(d) 96.764 m
(e) None of the above.

35. The horizontal distance of line QD is
(a) 96.559 m
(b) 65.456 m
(c) 63.077 m
(d) 96.764 m
(e) None of the above.

36. The bearing of line QD is
(a) 153° 54' 45"
(b) 151° 06' 05"
(c) 125° 58' 35"
(d) 331° 46' 05"
(e) None of the above.

29. Magnetic declination is defined as

(a) the horizontal angle from the grid north to the magnetic north.
(b) the horizontal angle from the true meridian to the magnetic meridian.
(c) the angle of inclination of the magnetic needle to the horizontal.
(d) the horizontal angle from the true north to the grid north.
(e) None of the above

30. Which of the following statements is/aretrue about the term *change point* as used in levelling?

I. A change point is a staff position at which two staff readings are taken.
II. At a change point, a foresight is taken first followed by a backsight.
III. At a change point, a backsight is taken first followed by a foresight.
IV. A change point is a levelling instrument position where a foresight is taken followed by a backsight.

(a) I and II only
(b) I and III only
(c) I and IV only
(d) II and IV only
(e) None of the above

During stadia tacheometric work on KNUST campus, a theodolite having a multiplying constant of 100 and an additive constant of 0 was correctly centred and levelled at a height of 1.450 m above a traverse station Q of reduced level 47.580 m. A levelling staff was held vertically at a traverse station P to provide a reference direction and then at the bases of two corners of a building labelled D and L in turn. The readings shown in Table 1 were taken.

Table 1: Tacheometric survey readings

Staff position	Staff reading (m)	Vertical circle reading	Horizontal circle reading
P	-	-	000° 00' 00"
Corner D	1.981, 1.497, 1.013	87° 08' 20"	102° 53' 45"
Corner L	1.773, 1.456, 1.142	91° 06' 20"	105° 02' 25"

Given that the coordinates of station Q are 800.000 m E, 650.000 m N, and the whole circle bearing of the line QP is 48° 52' 20", use the information in Table 1 to answer **questions 31 – 41**.

25. In a traverse survey, if the stations are occupied in a clockwise direction, which of the following statements is/are true?

(I) the interior angles are measured in a clockwise direction.
(II) the interior angles are measured in anticlockwise direction.
(III) the interior angles are measured in a horizontal plane.
(IV) the interior angles are measured in a vertical plane.

(a) I and II only
(b) I and III only
(c) II and III only
(d) II and IV
(e) None of the above

26. The bearing of a line AB is 175° 45' 27". What is the back-bearing of the line BA?

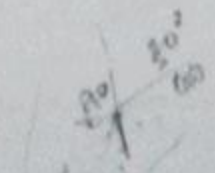
(a) 355° 45' 27"
(b) - 4° 14' 33"
(c) 175° 45' 27"
(d) 175°45' 27" ± 180°
(e) None of the above

27. The left-hand angle at a traverse station is

(a) the horizontal angle measured clockwise from the back station to the forward station.
(b) the vertical angle measured clockwise from the back station to the forward station.
(c) the horizontal angle measured clockwise from the forward station to the back station.
(d) the horizontal angle measured anticlockwise from the back station to the forward station.
(e) None of the above

28. The bearings of two lines PQ and PT are 060° 45' 27" and 290° 30' 42" respectively. Calculate the clockwise angle TPQ.

(a) 130° 14' 45"
(b) 229° 45' 15"
(c) 069° 29' 18"
(d) 351° 16' 09"
(e) None of the above



20. The horizontal length of section XY is
(a) 8.693 m.
(b) 17.926 m.
(c) 17.933 m.
(d) 8.707 m.
(e) None of the above

21. The standardization correction to be applied to measured length of section YB is
(a) - 0.095 m.
(b) + 0.095 m.
(c) - 0.002 m.
(d) + 0.002 m.
(e) None of the above

22. The horizontal length of section YB is
(a) 126.443 m.
(b) 126.253 m.
(c) 126.447 m.
(d) 126.257 m.
(e) None of the above

23. The horizontal angle measured clockwise from the meridian passing through P to line PQ could be
(I) the true bearing of the line PQ.
(II) the magnetic bearing of the line PQ.
(III) the back bearing of the line QP.

(a) I and II only
(b) I and III only
(c) II and III only
(d) All of the above
(e) None of the above

24. Find the quadrantal bearing of a line whose whole circle bearing is 157° 46' 25"
(a) N 12° 13' 35" W
(b) N 12° 46' 25" E
(c) S 22° 46' 25" W
(d) S 22° 13' 35" E
(e) None of the above

A line AB is measured in three sections AX, XY, and YB using a fibreglass tape of nominal length 20 m. The tape was read to the nearest 0.01 m and distances obtained for the three sections AX, XY, and YB were 79.95 m, 8.70 m, and 126.70 m, respectively.

There is a constant slope from A to X and from Y to B, while stepping was carried out between X and Y due to very steep ground. The reduced levels of points A, X, Y, and B are 37.52 m, 32.14 m, 16.87 m, and 20.21 m, respectively. Before measurement, the tape was measured against a reference steel tape and found to be 20.015 m.

Use this information to answer **questions 16 – 22**.

16. The standardisation correction to be applied to measured length of section AX is
    (a) - 0.060 m.
    (b) + 0.060 m.
    (c) - 0.015 m.
    (d) + 0.015 m.
    (e) None of the above.

17. The slope correction to be applied to the measured length of section AX is
    (a) - 0.189 m.
    (b) + 0.189 m.
    (c) - 0.378 m.
    (d) + 0.378 m.
    (e) None of the above.

18. The horizontal length of section AX is
    (a) 79.699 m.
    (b) 79.390 m.
    (c) 79.201 m.
    (d) 79.321 m.
    (e) None of the above.

19. The slope correction to be applied to the measured length of section XY is
    (a) zero because the ground between that section is very steep.
    (b) zero because stepping was used.
    (c) - 9.226 m.
    (d) + 9.226 m.
    (e) None of the above.

(a) I and II only
(b) I and III only
(c) II and III only
(d) All of the above
(e) None of the above

50-m and 70-m contour lines pass through two points A and B respectively. Assuming there is a constant slope between A and B and the horizontal distance between A and B is 120.000 m, use this information to answer questions 13and 14.

13. What is the slope distance between A and B?
(a) 112.655 m
(b) 121.655 m
(c) 116.322 m
(d) 124.000 m
(e) None of the above

14. Which of the following statements is not true?
(a) A is on a lower elevation than B.
(b) B is on a higher elevation than A.
(c) Reduced level of B is higher than that of A.
(d) The slope angle between A and B is 9° 27' 44.4"
(e) None of the above

15. Which of the following statements is/are true about a zenith angle?
(I) A zenith angle is between 0 and 180°.
(II) A zenith angle is measured in a vertical plane downward from an upward directed vertical line through the instrument.
(III) A zenith angle is measured in a vertical plane from a horizontal line upward or downward.

(a) I and II only
(b) I and III only
(c) II and III only
(d) All of the above
(e) None of the above

A distance of 100.00 m was measured between two points **P** and **Q** along a smooth slope of 1 in 5, **P** being on a lower elevation.

Use this information to answer questions **8 – 10**.

8. The slope angle is
   (a) 0.2°
   (b) 11.31°
   (c) 0.25°
   (d) 20°
   (e) None of the above

9. The horizontal distance between points **P** and **Q** is
   (a) 19.612 m.
   (b) 119.612 m.
   (c) 98.058 m.
   (d) 20.000 m.
   (e) None of the above

10. The difference in elevation between **P** and **Q** is
    (a) 19.612 m.
    (b) 20.000 m.
    (c) 1.942 m.
    (d) 18.058 m.
    (e) None of the above

11. A contour is defined as
    (a) a line joining points of equal size.
    (b) a line joining points of equal temperature.
    (c) a line joining points of equal pressure.
    (d) a line joining points of equal height.
    (e) None of the above

12. Which of the following characteristics is/are true of contour lines?
    (i) Contour lines are continuous lines within the areas they cover;
    (ii) Contours which are closer together indicate gentle slope and contours which are evenly spaced indicate steep slope;
    (iii) A contour line do not split into other contours or join any other contour line.

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4. Which of the following statements is/are **not**correct?

I. Accuracy is a measure of the absolute nearness of a measured quantity to its true value

II. Precision is the degree of consistency between measurements of the same quantity

III. The size of discrepancies in a data set is a measure of its precision

IV. Random errors are associated with the skill and vigilance of the surveyor

(a) I and II only
(b) II and IV only
(c) I and III only
(d) III and IV only
(e) None of the above

5. If 50 mm measured on a plan represents 50 m on the ground, the scale of the plan expressed in representative fraction is

(a) 1 : 50.
(b) 1 : 100.
(c) 1 : 1000.
(d) 1 : 2,500.
(e) None of the above.

6. Compute the area of a triangular plot measuring 60 m by 50 m by 40 m on the ground.

(a) 75.000 m^2
(b) 929.157 m^2
(c) 992.157 m^2
(d) 984,375.000 m^2
(e) None of the above

7. A plot of land measures 45 mm^2 on a plan of scale 1:2500. What is the equivalent area of the plot on the ground?

(a) 28,125.00 m^2
(b) 281.25 m^2
(c) 112,500.00 m^2
(d) 112.50 m^2
(e) None of the above

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

B. Sc. (Architecture) First Semester Examination, 2013

Second Year

December, 2013 **GE 183 PRINCIPLES OF LAND SURVEYING** Three (3) Hours

1. Which of the following statements is/are **not** correct?

   I. No measurement is exact.

   II. The exact error present in a measurement is always unknown.

   III. Systematic errors follow some physical laws and can be corrected for mathematically or by following proper field procedures.

   IV. Topographic surveys are made to determine the configuration (or relief) of the Earth's surface and to locate natural and artificial features on it.

   (a) I and II only
   (b) I and III only
   (c) I, II and III only
   (d) II, III and IV only
   (e) None of the above

2. Discrepancy is

   (a) the algebraic difference between two measurements of the same quantity.
   (b) the degree of consistency between measurements of the same quantity.
   (c) the measure of the absolute nearness of a measured quantity to its true value.
   (d) the spread of the measured values of a quantity.
   (e) None of the above

3. Which of the following statements is/are correct?

   I. Measurement is the application of a device or apparatus for the purpose of ascertaining an unknown quantity.

   II. Every measurement contains errors.

   III. Small errors occur less frequently than large ones.

   IV. An error is the difference between a measured quantity and its true value.

   (a) I and II only
   (b) II and III only
   (c) I and IV only
   (d) I, II and IV only
   (e) All of the above

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Step 7: Adjustment of the Latitudes and Depart

For any closed traverse, the linear misclosure must b
the traverse to close or balance the figure. The Con
the latitudes and departures of the traverse legs in
used.

$$\text{Correction in latitude for leg AB} = \frac{\text{-(total latit}}{\text{total tra}}$$

$$\text{Correction in departure for leg AB} = \frac{\text{-(total d}}{\text{tota}}$$

Step 8: Computation of rectangular coordinate

Northing N_B and Easting E_B coordinates of a forwa

$$N_B = N_A + \text{latitud}$$

8. Depending on the relative positions and elevations of two points A and B, the measured between them could be

   i. Horizontal distance
   ii. Slope distance
   iii. Vertical distance

   A. i & ii only
   B. i & iii only
   C. ii & iii only
   D. i, ii & iii only

9. Distances can be measured and set out

   i. directly using tapes
   ii. indirectly using Tacheometric techniques
   iii. Using EDM means

   A. i & ii only
   B. i & iii only
   C. ii & iii only
   D. i, ii & iii only

10. Which correction in distance measurement during a chain survey is always negative?

   A. Temperature correction
   B. Standardization
   C. Tension correction
   D. Slope correction
   E. None of the above

11. Systematic errors

   i. Vary in sign and magnitude
   ii. Remain even after all error types have been removed
   iii. Rarely occur

   A. i, ii, iii are all true
   B. Only i is true
   C. i and ii are true
   D. i, ii, and iii are all false

12. The varying effect of wind and temperature on a survey measurement can be classified as

   A. Human error
   B. Gross Error
   C. Systematic Error
   D. Random error
   E. None of the above

13. A 30m tape was used to measure a distance of 45.56m. The tape was later on found to ha actual length of 29.95m, what is the actual distance measured?

SECTION A

1. The internal angles of a close-loop anticlockwise traverse ABCDEA were observed with theodolite as follows: A = 89° 39′ 40″, B = 129° 18′ 50″, C = 83° 41′ 00″, D = 112° 38′ E = 124° 43′ 10″. The bearing of lin AE is fixed at 254° 58′ 00″ and the coordinates o (1000.00 m E, 500.00 m N).
The following horizontal distances were measured: AB = 120.39 m, BC = 232.71 m, CD = m, DE = 152.00 m, EA = 186.13 m. Calculate:
   (i) the angular misclosure of the traverse and adjust the angles
   (ii) the forward bearings of the traverse legs
   (iii) the coordinate differences for the traverse and adjust by the Bowditch method
   (iv) the fractional misclosure
   (V) the coordinates of stations B ,C ,D, and E
   (vi) the area of the traverse using the cross coordinate method.

2. During a stadia tacheometry work along part of a boundary of a site, a theodolite multiplying and additive constants of hundred and zero respectively was used. A leveli was held vertically at traverse stations B, C and D where stadia readings were taken on t as shown in the table below. The reduced level for point A is 153.200 m. Compute:

Inst. Stn.	Inst. Height(m)	Staff Stations (m)	Stadia readings (m)			Horizontal Circle reading
			Top	Mid	Lower	
A	1.460	B	1.684	1.200	0.716	320° 20′ 30″
		C	1.867	1.497	1.127	060° 40′ 30″
		D	1.888	1.420	0.952	180° 40′ 30″

   i. the horizontal distance of the perimeter ABCDA
   ii. the reduced levels of B,C and D
   iii. the total area of the polygon ABCDA
   v. the gradient of the line AC.

A. i, ii, iii,iv
B. ii, iii, iv, i
C. i, iii, ii, iv
D. i, iii, iv, ii
E. iv, i, iii, ii

27. Which of the following statements are correct?
i. Foresight and Backsight are recorded on the same line at a change point.
ii Foresight is the first reading at a setup.
iii Optical Square is used to set perpendiculars.

A. i and ii only
B. i and iii only
C. ii and Iii only
D. i, ii, iii only

28. Which of the following are **correct** about tilting level?
i. The telescope moves in the vertical plane only
ii. The telescope moves in the horizontal plane only
iii. The telescope moves both in the vertical and horizontal planes.
iv. The effect of the Tilting screw enables the telescope to move in the horizontal p
A. i, ii, iii and iv
B. i, ii, and iii only
C. ii and iv only
D. iii and iv only
E. None of the above.

29. Which of the following is a temporary adjustment of the Level instrument?
i. Elimination of parallax.
ii. Setting of the tripod
iii. Levelling of the instrument
iv. Elimination of zero error.
A. i, ii, iii and iv
B. i, ii, and iii only
C. i, ii, and iv only
D. ii, iii, and iv only
E. None of the above.

30. The following statements are related to Change points in levelling operation. Which of
are correct?
i. Backsight readings are observed before foresight readings at change points.
ii. It is the last station to be occupied by the staff man before the level instrume
position is changed.
iii. Change points should be stable

A. i and ii only
B. i and iii only
C. ii and iii only
D. i, ii, iii only
E. None of the above

SECTION B

1. Surveying practice involves which of the following:
   i. positioning of objects in space
   ii. positioning of physical features, structures and engineering works on the earth surf
   iii. Planning of the use, development and re-development of property
   A. i and ii only
   B. i and iii only
   C. ii and iii only
   D. i, ii and iii

2. The measured quantities in basic surveying are processed to determine the:
   i. configuration of the ground
   ii. lengths and directions of lines
   iii. areas of tract bounded by given lines

   A. i and ii only
   B. i and iii only
   C. ii and iii only
   D. i, ii and iii

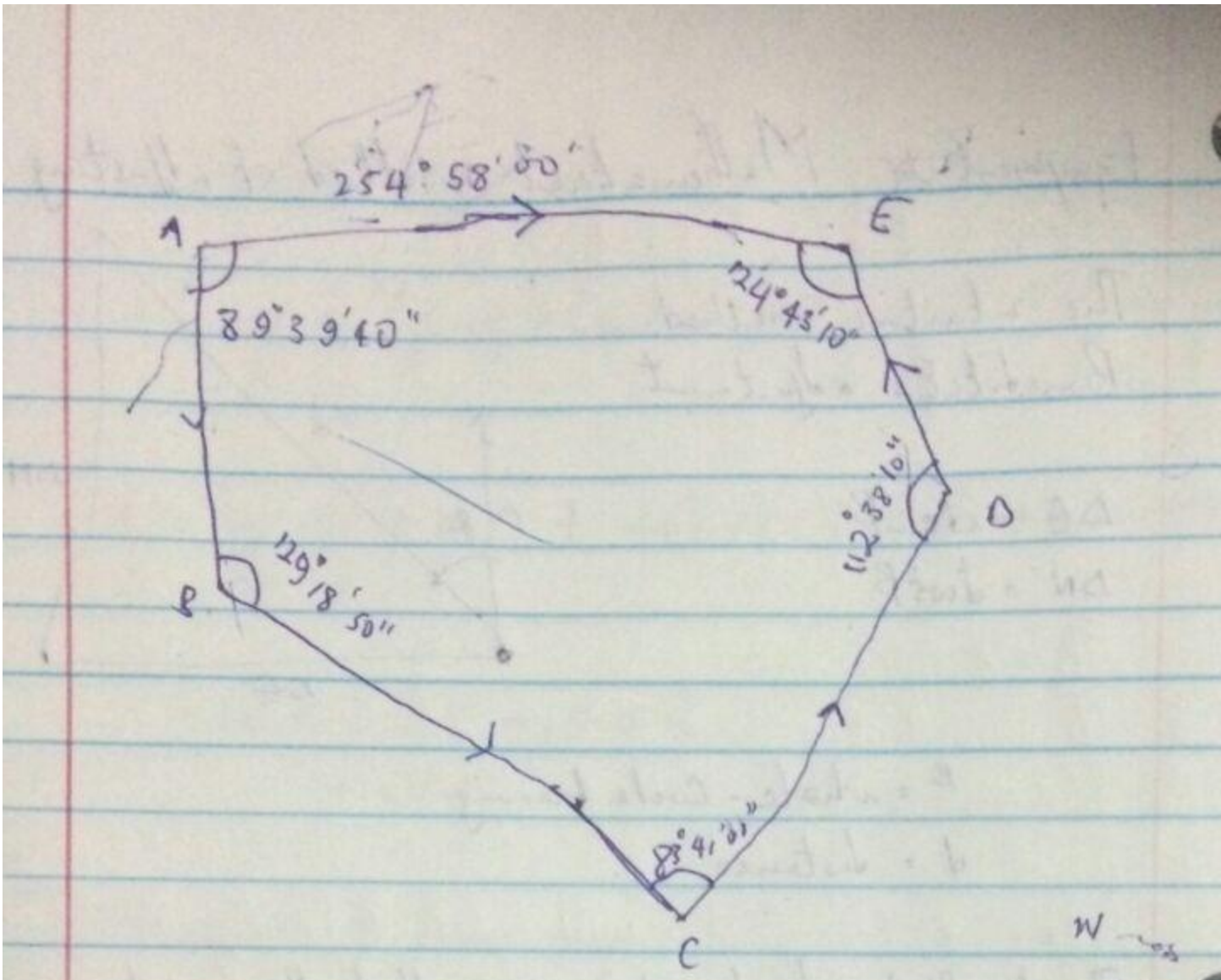
3. The type of surveying in which the curvature of the earth is taken into consideration is kn
   A. Geodetic Surveying
   B. Plane Surveying
   C. Hydrographic Surveying
   D. Engineering Surveying

4. In surveying the measurement of an angle and a distance with reference from a stati object is known as
   A. Trilateration
   B. Intersection
   C. Traversing
   D. Resection
   E. Triangulation

5. Topographic surveying is used to produce a map/plan which would contain features exce
   A. Natural/Artificial features
   B. Towns and villages
   C. Relief
   D. Very accurate boundary features

6. What name is given to the process in surveying where only linear measurement conditioned triangles are used?
   A. Trilateration
   B. Intersection
   C. Traversing
   D. Triangulation
   E. None of the above

... of obtaining horizontal distances directly on the field by dividing th



Bearing of Line AE = 254° 58' 00"

A (1000m E, 500.00m N)

	f. Bearing	L	ΔN.	Δ
AB	344° 37' 40"	120.39	116.08	-31
BC	293° 56' 30"	232.71	94.44	-21
CD	197° 37' 30"	168.30	-160.40	-50
DE	130° 15' 40"	152.00	-98.23	115
EA	74° 58' 0"	186.13	48.28	179

Part 2: ANSWER ALL QUESTIONS IN THIS PART

1. Explain why the residual gas saturation in the water-invaded zone in a wat reservoir is higher on the average than the corresponding residual oil satur drive/oil reservoir
2. An aquifer model requires input of aquifer properties which usually are un Explain how the needed aquifer parameters are obtained/calculated
3. Like most correlation models (e.g. Stone 1,Stone 2) Corey Correlation rel relative permeability modelling have been found to be valid for reservoirs characteristics. Name/state three of such characteristics.
4. i. State five reservoir characteristics which cause gas-cap expansion to rec ii. Of the five reservoir characteristics stated what do they all have in com
5. To predict the performance of a solution gas drive reservoir we require bo instantaneous gas-oil ratio equation and an equation to express the averag Derive the instantaneous gas-oil ratio equation and use the equation to ex shape of the producing GOR of a depletion type reservoir from a pressur bubble point to one significantly below the bubble point pressure. (20 ma
6. Capillary pressure data are obtained from core samples which represent the reservoir. Leverett derived a "J" Function using the Poiseuille equati flow:-to relate Capillary pressure (Pc), to Permeability (k), and Porosity function and comment on one of its limiting assumptions

$$q = \frac{\pi r^4 \Delta P}{8 \mu L}$$

7. Discuss any two of the primary drive mechanisms with appropriate dia
8. Discuss briefly with the aid of appropriate diagrams any one of the fol critical gas-condensate, Wet gas and Dry gas
9. Explain the following terms giving each their mathematical definition compressibility, Rock-Bulk compressibility and pore compressibility
10. Briefly describe the main differences in the flow regimes

38. Which of the following characteristics of contour lines are correct?

i. They are continuous lines
ii. They do not split/ join any other line except in cases of cliff
iii. They join places of the same height.

A. i and iii only
B. i and ii only
C. ii and iii only
D. i, ii and iii.

39. Offsets measured outside the boundary AD to an irregular boundary at a regular interval o from A, are as below. Calculate the area of the irregular figure using Simpson's rule

Distance from A (m)	0.0	30	60	90	120	150	180
Offsets (m)	0.0	3.7	4.9	4.2	2.8	3.6	0.0

A. 576.00 m^2
B. 567.00 m^2
C. 614.00 m^2
D. 641.00 m^2
E. None of the above

40. Offsets measured outside the boundary AD to an irregular boundary, are as below. Calcu area of the irregular boundary and the chain line AD using trapezoidal rule

Distance from A (m)	0.0	8	16	24	32	40	48	60	68
Offsets (m)	0.0	4.5	5.5	9.1	6.5	6.3	5.1	9.7	4.0

A. 446.40 m^2
B. 431.60 m^2
C. 414.00 m^2
D. 464.00 m^2
E. None of the above

C. Y. Asante
J. Ayer

Corr to ΔN	Corr to ΔE	ΔN Adjusted	ΔE Adjusted	N	E
		~~116.104~~	~~-31.885~~	500	1000
0.024	0.027	116.104	-31.885	616.104	968.117
0.046	0.051	94.486	-212.619	710.59	755.478
0.033	0.037	-160.367	-50.923	550.223	704.555
0.030	0.036	-98 2	116.024	452.023	820.579
0.037	0.041	48.317	179.801	500 34	1000.38
0.17	0.19				

$$\frac{1}{2} \times (2353857.463 - 2448763.192)$$

$$= 47452.86 m^2$$

Table 4.4: Computation of Latitudes and Departures ar

Stations

From (1)	W.C.B. (2)	Horizontal Distance (m) (3)	Latitudes (m)			
			ΔN (4),	Cor (5)	Cor. ΔN (6)	Δ
A	185° 11′ 00″	429.37	-427.61	0.02	-427.59	-1
B	100° 11′ 00″	656.54	-116.08	0.03	-116.05	6
C	50° 00′ 00″	301.83	194.01	0.01	194.02	2
D	00° 36′ 00″	287.40	287.38	0.01	287.39	
E	291° 05′ 40″	526.72	189.57	0.03	189.60	-4
F	250° 00′ 00″	372.47	-127.39	0.02	-127.37	-1
		Σ 2574	-0.12	+0.12	0	

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI
COLLEGE OF ENGINEERING
DEPARTMENT OF GEOMATIC ENGINEERING

B. Sc. (Petroleum Engineering), Third Year, 2015

GE 183 Principles of Surveying

Assignment 1

A piece of land is has been acquired by the College of Engineering wants to build a Hostel for the students in the faculty. You have been asked to provide the plan of the land for subsequent design, outline systematically the survey procedures you will adopt to produce the plan.

Assignment 2

The following data was obtained from a compass traverse. The bearings have been corrected for local attractions

Line	Bearing	Length
AB	29°30′	83.5
BC	100°45′	61.8
CD	146°30′	62.0
DE	242°00′	51.2
EA	278°45′	90.4

a) Plot the survey to a scale of 1:1000
b) Adjust the survey graphically by Bowditch's

Submission Date: **Latest by 27th November 2015**

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Dept of Geomatic Engineering